

OFFICIAL DOCUMENTATION · V1.1.0

CrossSwap

Cross-Chain Multi-Blockchain Exchange
with ChangeNOW Integration

LIVE DEMO

<https://cross-swap.blockshark.com>

DEMO CREDENTIALS

ADMIN

admin@demo.com

CUSTOMER

user@demo.com

PASSWORD

password!321

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Requirements

CrossSwap is a self-hosted Laravel 12 + Inertia + React application. It runs on any PHP-capable host — shared, VPS, container, or bare-metal — provided the runtime versions below are available.

Architecture at a glance

Storefront	React 19 + Inertia, served as a single-page app from Laravel.
Backend	Laravel 12 (PHP 8.2+) with queued jobs for status polling.
Database	SQLite by default; MySQL / MariaDB / PostgreSQL supported.
Exchange engine	ChangeNOW v2 partner API (non-custodial).
Cache & queues	Database driver works out of the box; Redis + Horizon recommended for production.
Mail	Any SMTP server, Mailgun, Postmark, SES, or log driver.

Server versions

Component	Minimum	Recommended
PHP	8.2	8.3 or 8.4
Composer	2.x	Latest stable
Web server	Apache 2.4 / Nginx 1.20	Nginx 1.24+
Database	SQLite 3.35 / MySQL 8.0 / MariaDB 10.6 / PostgreSQL 14	MySQL 8.0+ or PostgreSQL 16+
Node.js	not required at runtime	20+ only if rebuilding assets
Memory (RAM)	256 MB	512 MB+
Disk	250 MB free	1 GB+

Node is optional. The archive ships a fully built `public/build/` directory. You only need Node + npm if you want to recompile the front-end after editing Tailwind tokens or React source.

PHP extensions

The runtime needs the following PHP extensions. Most are bundled with the default PHP package on Ubuntu, Debian, RHEL, and almost every shared host.

`bcmath`, `ctype`, `curl`, `dom`, `fileinfo`, `json`, `mbstring`, `openssl`, `pcre`, `pdo` (plus one of `pdo_sqlite`, `pdo_mysql`, `pdo_pgsql`), `tokenizer`, `xml`, `zip`, `intl`, `sodium`, and either `gd` or `imagick` for image processing.

To install everything on a fresh Ubuntu/Debian box:

```
sudo apt update
sudo apt install -y php8.2-cli php8.2-fpm php8.2-mbstring php8.2-xml \
    php8.2-curl php8.2-zip php8.2-bcmath php8.2-intl php8.2-gd \
    php8.2-sqlite3 php8.2-mysql php8.2-pgsql php8.2-redis
```

Recommended optional extension: `redis` — enables Horizon-backed queues for background jobs (status sync, stuck-swap detection, recurring orders, mail send).

External services

- **ChangeNOW partner API key** — free, sign up at `changenow.io/affiliate`. Earnings post automatically to your partner balance.
- **Outbound HTTPS** to `api.changenow.io` on TCP/443.
- **SMTP credentials** (optional) for ticket replies, password resets, swap receipts. Configurable later in the admin.
- **OpenAI or Anthropic API key** (optional) for the AI chat widget. Falls back to the ticket portal if not configured.

Hosting checklist

Before running the installer, confirm:

- Document root points to `public/`, not the project root.
- URL rewriting enabled (`mod_rewrite` for Apache, `try_files` for Nginx).
- HTTPS available — Let's Encrypt via `certbot` is enough.
- `storage/` and `bootstrap/cache/` writable by the web server user.
- A cron entry that runs `php artisan schedule:run` every minute.
- A long-running queue worker via `php artisan queue:work` or `php artisan horizon`.

Installation

Step 1 — Upload

Extract the archive on your local machine, then upload the contents of `main_files/crossswap/` to your web hosting account. Use SFTP, SSH/rsync, or the file manager built into cPanel/Plesk.

Step 2 — Document root

Point your domain at the `public/` directory of the uploaded files. On cPanel/Plesk, edit the domain's "document root" setting. On a custom VPS, edit the nginx or Apache vhost so the root maps to `/path/to/crossswap/public`.

Example Nginx vhost

```
server {
    listen 80;
    server_name your-domain.com;
    root /var/www/crossswap/public;
    index index.php;

    location / { try_files $uri $uri/ /index.php?$query_string; }
    location ~ \.php$ {
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $realpath_root$fastcgi_script_name;
        include fastcgi_params;
    }
    location ~ /\.(!well-known).* { deny all; }
}
```

Step 3 — Permissions

Two directories need to be writable by the web server user (usually `www-data`):

```
chown -R www-data:www-data /path/to/crossswap
chmod -R 775 /path/to/crossswap/storage
chmod -R 775 /path/to/crossswap/bootstrap/cache
```

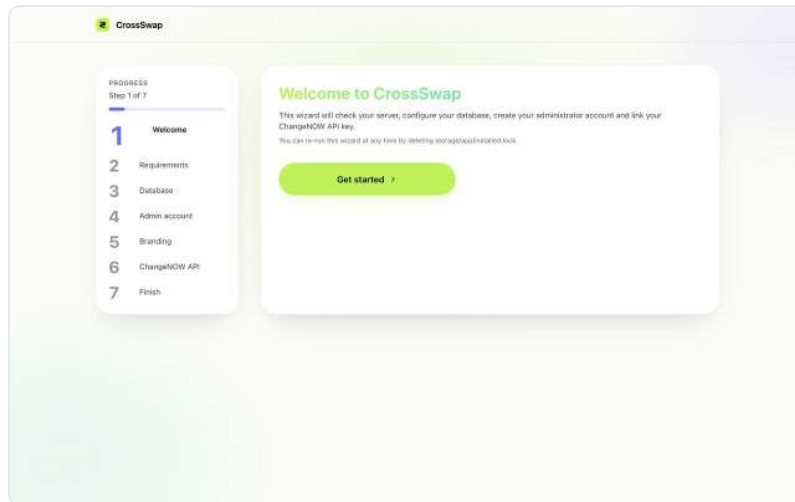
Step 4 — Web installer

Open `https://your-domain.com/install` in a browser. The seven-step wizard walks you through the rest. Each screen has a "Next" button at the bottom — the installer will not advance until the

current step is valid.

4.1 Welcome

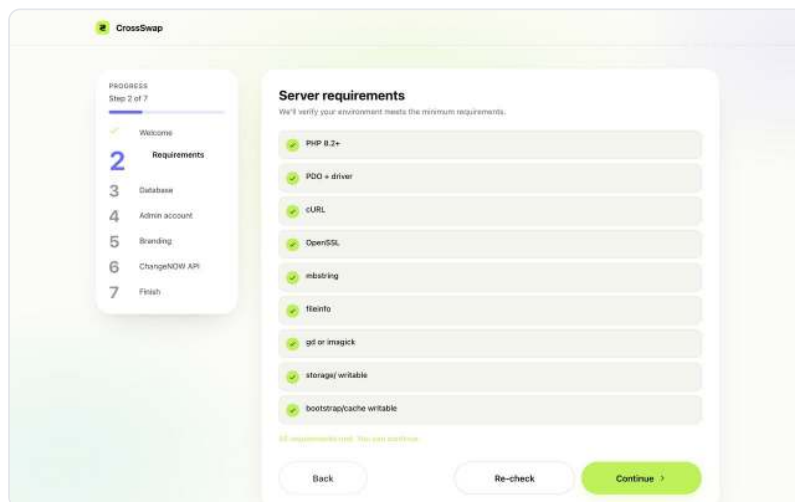
Introductory screen with a brief overview and a "Get started" button. No input required.



Step 01 — Welcome.

4.2 Requirements check

The installer audits PHP version, every required extension, and folder permissions on `storage/` and `bootstrap/cache/`. Each row is green or red. Fix any reds before continuing — you can refresh the page after each fix.



Step 02 — Requirements check.

4.3 Database

Pick a driver. SQLite is the zero-config default — the installer creates `database/database.sqlite` for you. For MySQL or PostgreSQL, enter host, port, database name, username, and password; the installer runs a test connection before moving on.

CrossSwap

PROGRESS
Step 3 of 7

- ✓ Welcome
- ✓ Requirements
- 3 Database**
- 4 Admin account
- 5 Branding
- 6 ChangeNOW API
- 7 Finish

Database connection
Choose your storage driver and credentials. The migrator will run automatically.

Driver
SQLite (file)

Database name
database/database.sqlite
Leave the path as database/database.sqlite to use the default file.

Back Test & migrate

Step 03 — Database.

4.4 Admin account

Create the first admin user — email, full name, password. This account is granted the `superadmin` role and can manage every other admin from **Admin** → **Users**.

CrossSwap

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Create your administrator
This account has full control of the admin panel. Pick a strong password.

Full name
Acme Crypto

Email address
you@your-domain.com

Password (12+ characters)

Confirm password

Back Create administrator

Step 04 — Admin account.

4.5 Branding

Upload a logo (PNG/SVG/JPG), set the brand name, tagline, and primary brand colour. All of these can be changed later under **Admin** → **Settings** → **Brand**.

The screenshot shows the 'Branding' step of the CrossSwap installation. On the left, a progress bar indicates the current step is 5 of 7. The main form area is titled 'Branding' with the subtitle 'Customize how the storefront presents itself.' It contains four input fields: 'Brand name' (filled with 'CrossSwap'), 'Tagline' (filled with 'Cross-chain swaps in 30 seconds, no account required'), 'Support email' (empty), and 'Default theme' (a dropdown menu with 'Dark' selected). At the bottom, there are 'Back' and 'Save branding' buttons.

Step 05 — Branding.

4.6 ChangeNOW API key

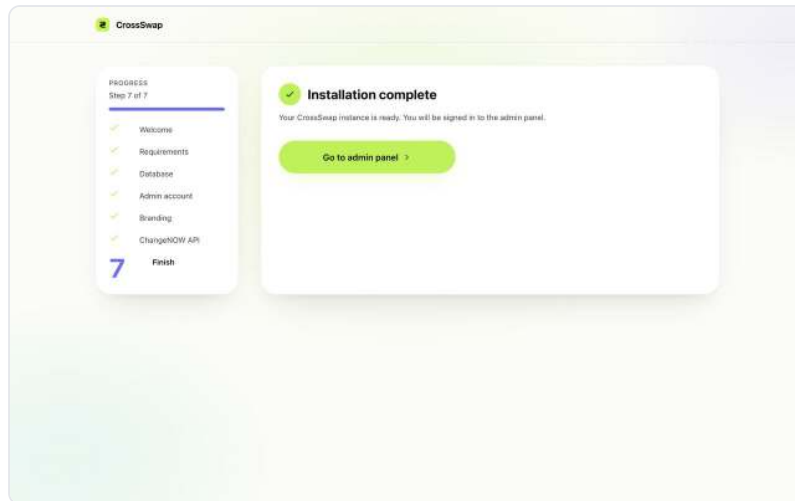
Paste your partner API key. The installer probes the ChangeNOW currencies endpoint and shows a green check on success or a clear error on failure (most often "invalid key" or "network blocked").

The screenshot shows the 'ChangeNOW API' step of the CrossSwap installation. On the left, the progress bar indicates the current step is 6 of 7. The main form area is titled 'ChangeNOW API' with the subtitle 'Paste your API key from your ChangeNOW partner account. We will verify it before saving.' It contains three input fields: 'API key' (filled with a masked string), 'Referral code (optional): optional-affiliate-tag' (empty), and 'Default exchange flow' (a radio button group with 'Standard (floating rate)' selected and 'Fixed-rate' unselected). Below these fields, there is a link 'Need a key? Visit your ChangeNOW partner dashboard.' and three buttons: 'Back', 'Test connection', and 'Save & continue'.

Step 06 — ChangeNOW API key.

4.7 Done

The finalise step writes `storage/app/installed.lock`, runs `migrate` + `db:seed`, optimises the app caches, and redirects you to the admin login. Sign in with the credentials from step 4.4.



Step 07 — Done.

The installer can be re-run safely. Delete `storage/app/installed.lock` and visit `/install` again. The DB schema migration is idempotent and existing rows are preserved.

Step 5 — Background workers (recommended)

CrossSwap uses queued jobs to poll swap status, flag stuck transactions, send receipts, and process recurring (DCA) orders. Configure a queue worker:

```
# With Redis + Horizon (recommended)
php artisan horizon

# Or with the database driver (no Redis needed)
php artisan queue:work --tries=3 --timeout=60
```

Cron entry

Add the Laravel scheduler to cron so prune jobs and recurring orders fire on time:

```
* * * * * cd /path/to/crossswap && php artisan schedule:run >> /dev/null 2>&1
```

Step 6 — Verify

- Visit `/` — the storefront swap card should load with live rates.
- Visit `/admin` — sign in with the credentials you created.
- Open **Admin** → **Dashboard** — KPIs and the 14-day chart should render.
- Place a test swap (e.g. BTC → ETH for a small amount) — it should appear in **Admin** → **Transactions**.

Configuration

All day-to-day configuration lives in the admin panel under **Admin** → **Settings**. Each tab persists immediately to the `settings` table and is read back into the running application without a deploy or cache flush.

Where things live

<code>.env</code>	Runtime secrets — DB credentials, app key, ChangeNOW API key, SMTP password. Never commit this file.
<code>config/</code>	Default values for everything. Read once at boot; override via <code>.env</code> or the admin.
<code>lang/</code>	One folder per locale (<code>en</code> , <code>es</code> , <code>de</code> , <code>fr</code>). Flat PHP arrays.
<code>storage/app/public/</code>	Uploaded brand logo + content media. Symlinked to <code>public/storage</code> by <code>storage:link</code> .
<code>storage/logs/</code>	Application logs. Tail <code>laravel.log</code> when debugging.
<code>database/database.sqlite</code>	Default SQLite file (only when DB driver is sqlite).

Brand

- **Logo** — drag-and-drop PNG/SVG/JPG. Stored in `storage/app/public/` and linked via `php artisan storage:link`.
- **Brand name** — appears in nav, footer, emails, and the document `<title>`.
- **Tagline** — homepage subtitle and the Open Graph description.
- **Social links** — Twitter/X, Telegram, Discord, GitHub. Hidden in the footer when blank.
- **Legal URLs** — privacy, terms, cookies. Either external URLs, or use the bundled Pages CMS to host them on the same domain.

Theme

Pick light or dark as the default and override the brand colour from the swatch palette. For colour tuning beyond the admin UI, edit the `@theme` block at the top of `resources/css/app.css` and rebuild with `npm run build`. The archive ships a pre-built bundle so Node + npm are only needed when you change tokens or React source.

ChangeNOW API

Set under **Admin** → **Settings** → **API**. The page includes a **Test connection** button that calls the ChangeNOW currencies endpoint and reports the response. If the API is unreachable or the key is invalid, the storefront falls back to a static featured-pair list and surfaces a friendly error in the swap card.

Common `.env` overrides:

```
CHANGENOW_API_KEY=your-partner-key
CHANGENOW_BASE_URL=https://api.changenow.io/v2
CHANGENOW_FLOW=standard          # or "fixed-rate"
CHANGENOW_RATE_TYPE=expected
CHANGENOW_CACHE_TTL=120          # seconds
```

SMTP / Mail

Configure under **Admin** → **Settings** → **Mail**. The **Send test email** button at the bottom of the form delivers a marked test message to a target address so you can validate credentials without leaving the admin. The same form also lets you set the "From" name and address used on outbound mail.

Currency picker

Admin → **Settings** → **Currencies** shows every coin returned by ChangeNOW with a search box and two per-coin toggles:

- **Featured** — promotes a coin into the homepage "Popular pairs" rail.
- **Blacklist** — hides a coin from the storefront entirely (still visible in the admin).

Content CMS

The four marketing blocks below the swap card — **Highlights**, **Reviews**, **Stats counter**, and **How it works** — are admin-editable per locale at **Admin** → **Content**. Each block ships a sensible default so a clean install renders out of the box; the defaults are replaced the moment an admin saves an override.

Pages (multi-locale)

Privacy, terms, cookies, about, and any other long-form page is a record under **Admin** → **Pages**. Each page can be translated into the active locales (per-locale tabs at the top of the editor) using a built-in tabbed Write / Preview Markdown editor. Header/footer visibility and sort order are admin-controlled.

Public URL pattern: `/p/{slug}`.

Translations

The shipped archive includes four locales — English, Spanish, German, French — each at full key parity (1134 keys). Translation files live at `lang/<locale>/site.php` as flat PHP arrays. To add a new locale:

1. Copy `lang/en/site.php` to `lang/<new-locale>/site.php`.
2. Translate each value, leaving the keys intact.
3. Add the locale code to the `supported_locales` array in `config/app.php`.
4. Run `php artisan optimize:clear` to refresh the cache.

To verify all locales are at the same key count:

```
php -r "\$flat = function(\$a,\$p='') use (&\$flat){\$o=[];foreach(\$a as \$k=>\$v){\$key=\$p===' '?\$k:\$p.\$k;if(is_array(\$v))\$o+=\$flat(\$v,\$key);else \$o[\$key]=\$v;}return \$o};";
foreach(['en','es','de','fr'] as \$l) echo "\$l: ".count(\$flat(include "lang/\$l/site.php")).PHP_EOL;"
```

Updating CrossSwap

When a new release is published on CodeCanyon:

1. Back up `.env` and the database.
2. Download the new ZIP and extract.
3. Upload the new `main_files/crossswap/` contents on top of the existing install. Keep your `.env` and the `storage/` tree intact.
4. Run `php artisan migrate --force`.
5. Run `php artisan optimize:clear`.
6. Restart your queue worker (Horizon or `queue:work`).

Common issues

- **419 PAGE EXPIRED** — usually a stale cookie. Clear browser cookies for the domain and try again. If it persists, run `php artisan optimize:clear`.
- **500 Server Error** — check `storage/logs/laravel.log`; the most common cause is missing write permissions on `storage/`.
- **ChangeNOW returned 401** — invalid API key. Re-paste it in **Admin** → **Settings** → **API**.
- **Mixed content over HTTPS** — set `APP_URL=https://your-domain.com` in `.env` and run `php artisan config:cache`.